Statistics Tutorial

What is the mass of an electron? How tall is an adult human being? Are left-handed people faster than right-handed people? Even if you have the appropriate tools (scales, rulers, clocks, etc.) to make measurements that address such questions, there are some inherent problems in translating measurements into answers.

One problem is that measurement equipment is never perfect. If you measure the mass of an electron over and over again as precisely as you can, you will probably get a range of results. Given the values you obtain, how well do you know its real mass?

A second problem is that the question you're asking may not have a definitive answer. Adult humans vary in height. Fortunately, those heights conform to a distribution. There are many more people of average height than people who are either very short or very tall. How can you efficiently and accurately describe this distribution?

A third problem is that it is commonly possible to make only a limited number of measurements. You would like to learn something about the total distribution of adult heights, but time and available resources allow you to sample only a small subset of people. Were you to measure the heights of 1,000 adults, how well would your data reflect the height distribution of the world's 5 billion adults?

Related problems occur when you try to compare two or more groups. On measuring the top running speed of 100 left-handed people and 100 right-handed people, you might find that on average the left-handers are faster. But some right-handers turn out to be faster than the majority of left-handers, and some left-handers slower than the majority of right-handers. Given the limited number of measurements and the fact that there is an overlap in the speeds for individuals from each group, when is it possible to conclude that a real difference exists between the two groups as a whole?

In science, all of these problems are addressed using statistics. Statistics provides tools to characterize measurements (means, standard deviations), compare them with the larger population (standard errors, confidence intervals), and discern differences between groups (statistical significance, p-values).

Means and Standard Deviations

In what follows it is assumed that you have made N number of measurements, sampled randomly from a large population, with measured values $X_1, X_2, X_3, \ldots X_N$.

The most familiar and common way of characterizing measurements is their **mean** or average, signified by $\overline{X}$. The mean is calculated by summing the measured values and dividing by the number of measurements:

$$\overline{\mathrm{X}} = \frac{1}{\mathrm{N}} \sum_{i=1}^{\mathrm{N}} \mathrm{X}_i \tag{1}$$

While the mean is a useful description, it doesn't tell you anything about how spread out the individual measurements are. For example, suppose you measure the heights of the students in two different Frontiers seminars. Both seminars turn out to have a mean height of 5 feet 7 inches, but the first seminar has students ranging from 4' 7'' to 6' 7'', while the second seminar ranges from 5' 2'' to 6'1''. You would describe these measurements by saying that while the means for the two seminars are the same, the first seminar has more variability.

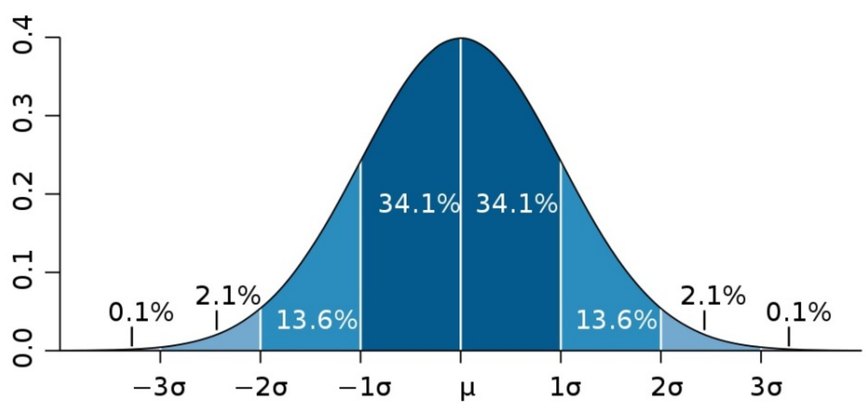
To characterize this variability we want a number that represents how far measurements typically deviate from the mean. If you calculate the differences between each measurement and the mean, you will get both positive and negative values, which will cancel each other when added up. To get around this difficulty we take the square of the difference, which is always positive, and calculate the square root of the average of the squared differences. This calculation tells us the **standard deviation** (SD) of the measurements from the mean:

$$\mathrm{SD} = \sqrt{\frac{1}{\mathrm{N}-1} \sum_{i=1}^{N} (\mathrm{X}_i - \overline{\mathrm{X}})^2} \tag{2}$$

While the formula may look complicated, the SD is simply a measure of how spread out the measurements are. In the example above, the heights of the first seminar covered a wider range than the heights of the second, so the SD of the heights in the first seminar was larger.

Normal Distributions

Typically, as each new measurement is made, the mean and SD change, but as the total number of measurements becomes large, the measured values will approach the distribution of the total population being examined (or some ideal of an infinite number of measurements). Often this total distribution is approximated by a curve called a **normal distribution**, also known as a bell curve or Gaussian.

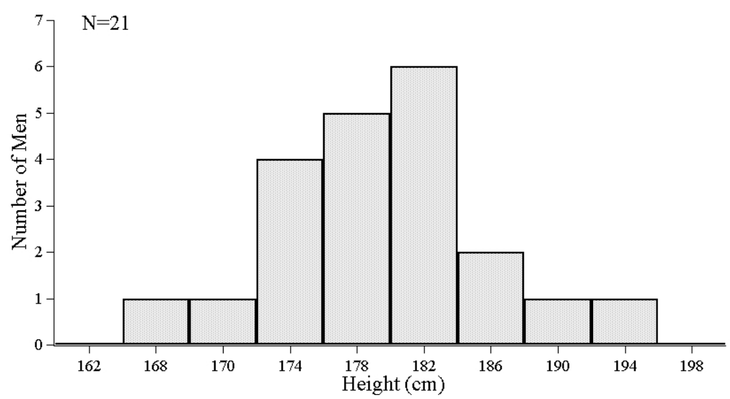


The mean value of this distribution is denoted by μ and its standard deviation by σ. For large enough N the measured mean, $\overline{X}$, and SD can approximate these idealized values. While the normal distribution is a mathematical idealization, it is a reasonable model for many types of total distributions. However, one should be aware that in many cases there are superior models. For example the distribution of the heights of entering Columbia College students is more accurately modeled as the sum of two normal distributions with different means – one for the heights of men and one for the heights of women.

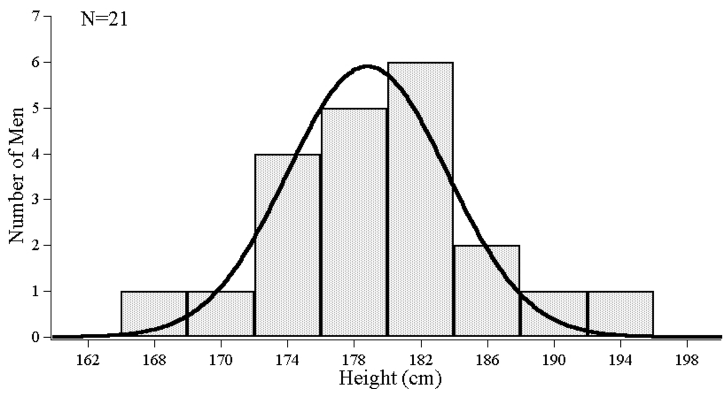
The great advantage of the normal distribution model is that it allows us to use the machinery of statistics to make statements about how probable the outcomes of measurements are based on measurements you've already made. The normal distribution above shows that only 34.1% of the total population falls between the mean and the mean plus 1 SD and only 0.1% is less than 3 SD below or above the mean. As an example, suppose you want to predict the percentage of CC women over 183 cm (6 feet) tall. You measure the heights of 25 CC women (perhaps none of them turning out to be over 183 cm) and find your measurements have an average value of 167 cm, and a SD of 8 cm. If the women you've measured are representative of all CC women, you can use $\overline{X}$ and SD to estimate μ and σ. The above graph shows that only around 2% of the total population should have a height greater than 183 cm ($\mu + 2\sigma$). If there were 2,000 CC women, you would then expect around 40 of them to be 6 feet or more.

Estimating Means and Standard Deviations from Graphs

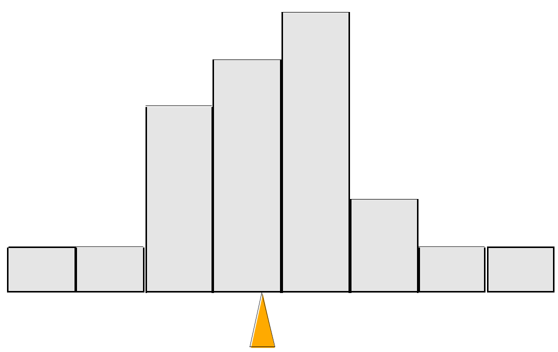
Given a set of measurements, you can use Equations (1) and (2) to calculate $\overline{X}$ and SD precisely. However, it is often more expedient and nearly as precise to estimate these values using graphs. Take the following histogram of the heights of 21 men:



What is the average height of these men? Because the graph is fairly symmetric, it should be close to the middle of the data, one might initially guess the middle of the 3 highest bars at 178. However, since the bar at 182 is larger than the one at 174 and the bar at 186 larger than the one at 170, the mean is probably a bit to the right of 178. Another way of visualizing the mean is to approximate the data by drawing or imagining a normal distribution that fits the histogram,

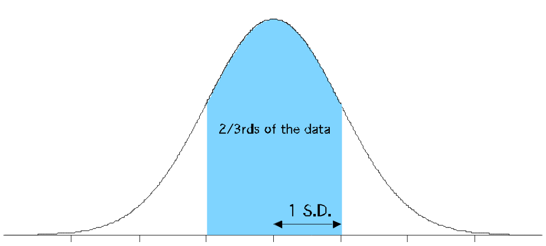


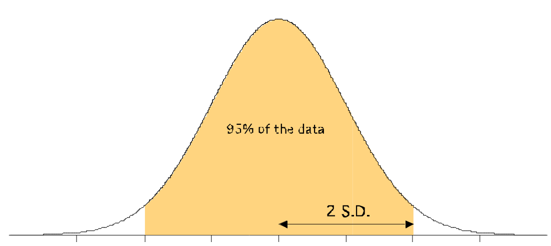
which again puts the mean (the peak of the curve) a bit to the right of 178. One can also imagine the whole graph as a solid object, and picture balancing it on a single point. The point that balances the graph is the mean:



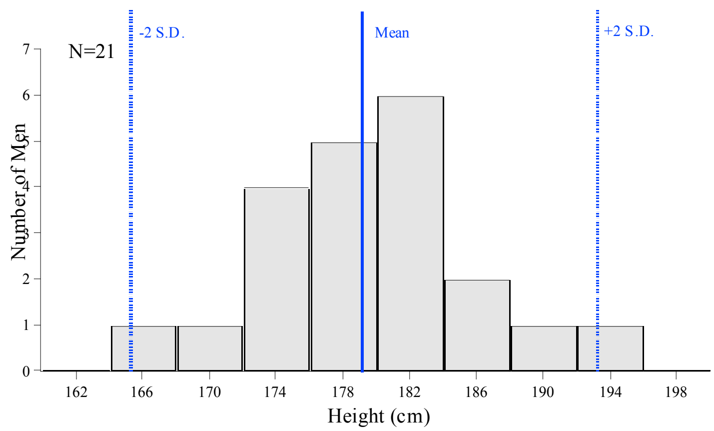
This visualization technique is especially useful for estimating means from histograms with asymmetric or skewed data. For these data, a good visual estimate of $\bar{X}$ is about 179 cm.

Once $\bar{X}$ has been established, the SD can also be estimated from the graph. For a normal distribution, the region from 1 SD below the mean to 1 SD above the mean contains about 2/3 of the data, and the region from 2 S.D below the mean to 2 SD above the mean contains about 95% of the data:

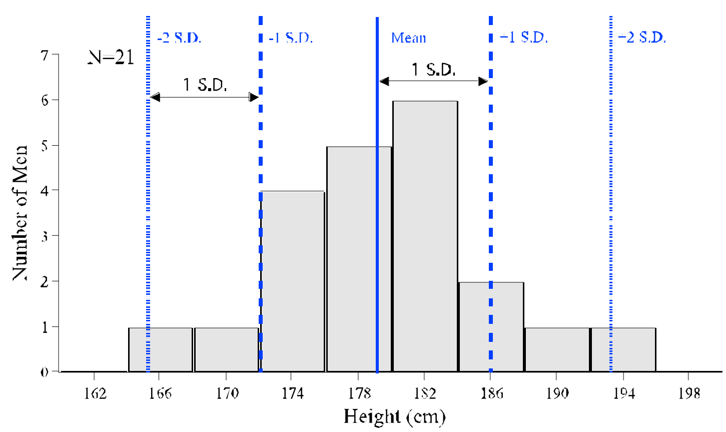




First, we estimate the region from $\overline{X}$ - 2 SD to $\overline{X}$ + 2 SD. In this example, almost all of the data lie in the range between 165 cm and 193 cm, so we can draw estimates for the mean and ± 2 SD as follows:



Then we can add another set of lines halfway in from 2 SD, and check that the resulting range for the mean ± 1 SD contains about 2/3rds of the data:



Here, the shaded area between the two dashed lines accounts for about 2/3 of the total. Similarly, the range between the two outside dotted lines contains about 95% of the data. Our estimate for the SD is then the distance between any two of these lines, that is, about 7 cm.

We've now estimated our $\overline{X}$ to be around 179 cm and our SD to be around 7 cm. Using formula (3) we can calculate an estimate for the SE provided we know the number of measurements made. In any scientific article this information (denoted N) is usually given somewhere, often on the graph or in the figure caption. In this case N = 21, so SE ≈ 7 cm/√21 = 1.5 cm. The values for $\overline{X}$, SD, and SE calculated directly from the data used to make this histogram are 178.8 cm, 6.8 cm, and 1.5 cm. Our estimated values are not perfect, but they are reasonably close.

Standard Errors

The $\overline{X}$ and SD measured from our sample of size N are estimates for the true mean and standard deviation (μ and σ) of the entire population. In order for these estimates to be useful, we need an idea of the error in our measured mean compared to the true mean of the population. Intuitively, the mean of our measurements should get closer to that of the whole population as more measurements are made. In addition, since very few measurements will fall more than a few standard deviations away from the mean, the smaller the standard deviation of the total population the more accurate your estimate of the mean. These considerations allow us to calculate an estimate for the **standard error of the mean**, SE

$$\mathrm{SE} \approx \frac{\mathrm{SD}}{\sqrt{\mathrm{N}}} \tag{3}$$

The SE gives us an idea of how close your measured mean is to the mean of the whole population or the mean in the ideal limit of infinitely many measurements. For example, in the measurement of CC women's heights SD = 8 cm and N = 25. So SE ≈ 8 cm/√25 = 1.6 cm. It is more likely than not that the mean height of all CC women falls within 1 SE, or 1.6 cm, of the measured mean of 167 cm, and the mean of the population as determined from this measurement could fairly be reported as 167 cm ± 1.6 cm. As you make more measurements, the SD will probably not change very much, but the increasing N will reduce the SE, allowing you to report the mean of the whole population with a smaller error.

It is important to note that the mean ($\overline{X}$), standard deviation (SD), and standard error of the mean (SE) all have the same units. In our example, we measure heights in centimeters, so all three quantities have units of centimeters. All three values are calculated from our sample of N measurements (randomly selected from a larger population), but while $\overline{X}$ and SD estimate properties of the entire population (the true mean and standard deviation of the population), the SE estimates a property of our measurement specifically, that is, how close our measurement of the mean is to the true mean.

Statistical Significance, Confidence Intervals, and p-values

How confident can you be that your measured mean ($\overline{X}$) is close to the mean of the whole population? This question leads to the concept of a **confidence interval**. For example, the interval from ($\overline{X}$ − 1 SE) to ($\overline{X}$ + 1 S. E.) is a 68% confidence interval; there is a 68% chance that the mean of the whole population is in this range. Scientists most commonly speak of 95% confidence intervals, which we will approximate as extending from 2 SE's below the mean to 2 SE's above.

A principal use of confidence intervals is in determining whether two groups are different with respect to some measure. For example, are CC men, on average, taller than CC women? Suppose you measure the heights of 9 CC men, finding an $\overline{X}$ of 178 cm and an SD of 9 cm. The mean height of the men you measured is greater than the mean height of the women you measured (167 cm), but how sure can you be that this is true for all of Columbia College? How can you determine whether or not this difference in means occurred only by chance in your limited number of measurements?

Here, confidence intervals can help. Since the mean of the women's heights had an SE = 1.6 cm, the 68% confidence interval for the female mean is 165.4 cm to 168.6 cm (that is, $\bar{X} \pm 1$ SE), and the 95% confidence interval for the female mean is about 163.8 cm to 170.2 cm ($\bar{X} \pm 2$ SE). So while we know our measured mean is not exact, we also know it is unlikely that the true mean is greater than 170.2 cm. For the men, the SE ≈ 9 cm/$\sqrt{9} = 3$ cm. So the 95% confidence interval is about 172 cm to 184 cm, and it is unlikely that the true male mean is less than 172 cm. Although the difference in mean male and female heights was measured only in a small portion of the population, by calculating the 95% confidence intervals of the two means we can say that the difference is **statistically significant**.

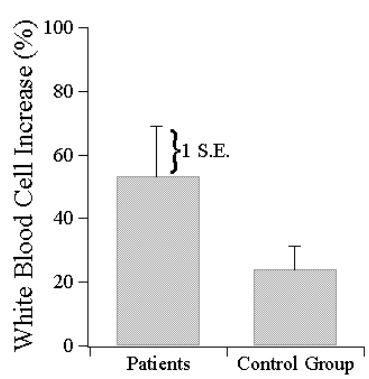
The criterion for what qualifies as a statistically significant difference in the mean values of two groups varies depending on scientific disciplines, but often involves using SE's and confidence intervals. In Frontiers of Science we will adopt the following criterion as an approximation:

The means of two groups of measurements differ by a statistically significant amount if and only if the ranges from 2 SE's below the mean to 2 SE's above the mean for each measurement do not overlap.

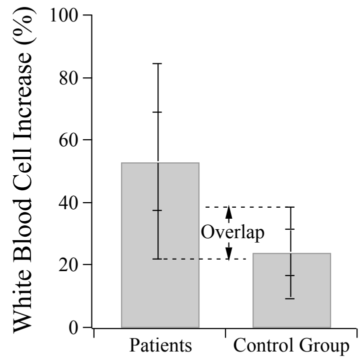
In scientific articles the **p-value** is the most frequently encountered parameter quantifying whether observed differences are statistically significant. Roughly speaking, the p-value is the probability that the observed difference could have happened by random chance. A p-value of 0.002 means that if there really were no difference between the groups measured, there is a 0.2% chance of obtaining data with at least as extreme a difference as is actually observed. The lower the p-value, the more likely the observed difference reflects a real difference in the underlying populations. In many fields, a p-value of less than 0.05 means the difference qualifies as statistically significant.

Estimating Statistical Significance from Graphs

Determining whether a difference in means is statistically significant can be straightforward using graphs. Suppose that two groups are suffering from a disease that suppresses the production of white blood cells. One group is given a new experimental treatment ("patients"); the other (the "control group") is given the standard treatment. After treatment the increase in the percentage of white blood cells for all individuals in each group is measured:



The count for the patient group has increased by around 55% on average, while the control group has seen only a 25% increase. Does this mean that the new treatment has produced a statistically significant benefit over the standard one? Our criterion for determining statistical significance requires an estimate of the SE for each group. In this case 1 SE is shown as an extended line on each of the bars.* The means shown on the bar graph can be extended 2 SE in each direction to see if these ranges overlap.



In this case the ranges do overlap. So even though the mean increase in white blood cells for the group given the experimental treatment is fairly large, we cannot say that the treatment has produced statistically significant improvements. This conclusion does not mean that the treatment is necessarily ineffective, only that this data alone not provide a compelling case in favor of the treatment. As mentioned above, what constitutes a "convincing case" varies from discipline to discipline, and depends largely on how feasible it is to conduct a large number of measurements. Our 2 SE non-overlap criterion is a bit more stringent than what is typically used in biological studies, but would be considered not stringent enough in the particle physics community.

Measurement Error: Random and Systematic

No measurement is perfect. Errors in measured quantities fall into two broad categories: random errors and systematic errors.

Random errors occur due to variability in measurements. In a careful measurement of nearly any quantity, the obtained result will be either greater than or less than the "true" value, by a small, random amount. Sometimes the result will be too high, sometimes too low, and as a result, repeated measurements of the same quantity will give a set of results scattered around the true value. Sources of random error may include fluctuation in the measurement apparatus itself (e.g., electrical noise in a digital multimeter), variability in the interaction of the measurement with the environment (e.g., air currents in a room changing the reading on a scale), or deviation in a human experimenter's interpretation and recording of results. A measurement with only a small amount of random error is said to high have precision.

Statistical tools can help describe and minimize the impacts of random error in a measurement. Since the results are scattered fairly evenly around the true value, the mean value of the measurements is not affected much by random errors, as long as enough measurements have been taken. Random errors do, however, increase the variability of the measurements and therefore increase both the standard deviation

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* On some graphs this error bar may represent 2 SE, 1 SD, or another value. The text or caption accompanying the graph must always be read carefully to determine its meaning. These statistical measures may also be provided in the text of an article rather than displayed on the graph.

(SD) and the standard error of the mean (SE). Increasing the total number of measurements reduces the contribution of random errors to the SE.[†]

Systematic errors represent a bias in the measurement, resulting in measured values that consistently fall either above or below the true value. Systematic errors can occur in many different ways. The bias may be inherent in the design or setup of the experiment. Asking a group of (stereotypical) fishermen to report the size of the fish they caught may result in a systematic over reporting of fish size. Likewise, measuring the fish of only those who volunteer to have their catches measured may result in a sampling bias, resulting in a systematically inflated measurement of a typical catch. Alternatively, the bias may be inherent in the measurement apparatus itself. Asking fishermen to measure their catches with the same faulty ruler may result in a systematic underreporting of size. A systematic error can also occur in the analysis of accurately measured data. Converting the fish length from inches to centimeters with the wrong unit conversion factor would result in a consistently incorrect value.

Systematic errors, particularly those with their roots in the design of the experiment itself, are the most difficult errors to identify and minimize. Since systematic errors bias all measurements in the same direction, they change the mean value of the measurements, by an amount that cannot be quantified using the standard error of the mean or other statistical techniques. A measurement with large systematic errors is said to have poor accuracy.

[†] Recall that for normal distributions, $SE = SD/\sqrt{N}$, where N is the number of measurements.